

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8CJ406 / MAT8MN406			
Course Title	BASIC MEASURE THEORY			
Type of Course	Major			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	3	2	75
Pre-requisites	1. Fundamental Mathematics Concepts: Set, Functions, Logic 2. Real Analysis			
Course Summary	This course familiarises students with the Lebesgue Measure on the real line and how it enables the construction of a theory of integration that does away with many of the drawbacks of Riemann integration.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and explain the concepts of Lebesgue measure, including outer measure, measurable sets, and properties such as countable additivity and the Borel-Cantelli Lemma.	U	C	Internal Exam/ Assignment/ Seminar/ Viva /Report/ End Sem Exam
CO2	Apply theorems related to Lebesgue measurable functions, including Littlewood's Three Principles, Egoroff's, and Lusin's Theorems, to analyse function behaviour and approximations.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva /Report/ End Sem Exam
CO3	Evaluate and integrate functions using the Lebesgue integral, understanding its differences from the Riemann integral and applying it to bounded and non-negative measurable functions.	E	F	Internal Exam/ Assignment/ Seminar/ Viva /Report/ End Sem Exam
	* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create(C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)			

Detailed Syllabus:

Text book	Real Analysis, H. L. Royden & P. M. Fitzpatrick, 4 th Edition, Prentice Hall of India, 2000			
Module	Unit	Content	Hrs (45+30)	Ext. Marks (70)
I	Chapters 0, 1, 2: The Lebesgue Measure		15	Min.15
	1	Preliminaries On Sets, Mappings & Relations (Review only)		
	2	Chapter 1: The Real Numbers: Sets, Sequences & Functions (Proofs of results included in Practicum)		
	3	2.1 Introduction – Measure as a set function		
	4	2.2 Lebesgue Outer Measure		
	5	2.3 The σ –Algebra of Lebesgue Measurable Sets		
	6	2.4 Outer & Inner Approximation of Lebesgue Measurable Sets		
	7	2.5 Countable Additivity, Continuity & the Borel-Cantelli Lemma		
	8	2.6 Non-Measurable Sets		
II	Chapter 3: Lebesgue Measurable Functions		8	Min.15
	10	3.1 Sums, Products & Compositions		
	11	3.2 Sequential Pointwise Limits & Simple Approximation		
	12	3.3 Littlewood’s Three Principles, Egoroff’s & Lusin’s Theorems		
III	Chapter 4: The Lebesgue Integral		12	Min.20
	13	4.1 The Riemann Integral		
	14	4.2 Lebesgue Integral of Bounded Measurable Function Over a Set of Finite Measure.		
	15	4.3 Lebesgue Integral of a Non-negative Measurable Function.		
	16	4.4 The General Lebesgue Integral		
	17	4.5 Countable Additivity & Continuity of Integration (proofs included in practicum)		
	18	4.6 Uniform Integrability: The Vitali Convergence Theorem (proofs included in Practicum)		
IV	Chapter 5: Differentiation & Lebesgue Integration		10	Min.10
	19	6.1 Continuity of Monotone Functions.		
	20	6.2 Differentiability of Monotone Functions: Lebesgue’s Theorem		
	21	6.3 Functions of Bounded Variation: Jordan’s Theorem		
	22	6.4 Absolutely Continuous Functions (Proof of Theorem 9 is optional)		
	23	6.5 Integrating Derivatives: Differentiating Indefinite Integrals		
V	Practicum: The goal is for the students to learn the following selected topics in 10 practicum sessions of three hours each via self-study and group activities. The lecturer may assist by running group discussions and supervising class seminars and referring library books for self-study and note preparations.		30	
	1	Proofs in Chapter 1: The Real Numbers		
	2	Section 2.7 - The Cantor Set & the Cantor-Lebesgue Function		
	3	Proofs in Section 4.5		
	4	Proofs in Section 4.6		

5	5.1: Uniform Integrability & Tightness		
6	5.2: Convergence in Measure		
7	5.3: Characterizations of Riemann & Lebesgue Integrability		
8	7.1: Normed Linear Spaces		
9	7.2: Inequalities		
10	7.3: Riesz-Fischer Theorem		

References

1. R. G. Bartle, Wiley, The Elements of Integration & Lebesgue Measure, 1995..
2. G. de Barra, Measure Theory & Integration, New Age International Publications, 1981.
3. David M. Bressoud, A Radical Approach to Lebesgue's Theory of Integration (ARALTI), Cambridge University Press, 2008.
4. P. R. Halmos, Measure Theory, GTM, Springer-Verlag
5. Walter Rudin, Principles of Mathematical Analysis, 3rd Edition, Tata McGraw Hill Inc., 1976.
6. Walter Rudin, Real & Complex Analysis, 3rd Edition, McGraw Hill Inc., 1987.

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	0	0	3	0	2	1	3	0	2
CO 2	2	2	0	0	3	0	3	2	3	0	3
CO 3	1	0	3	0	3	0	3	1	3	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓